

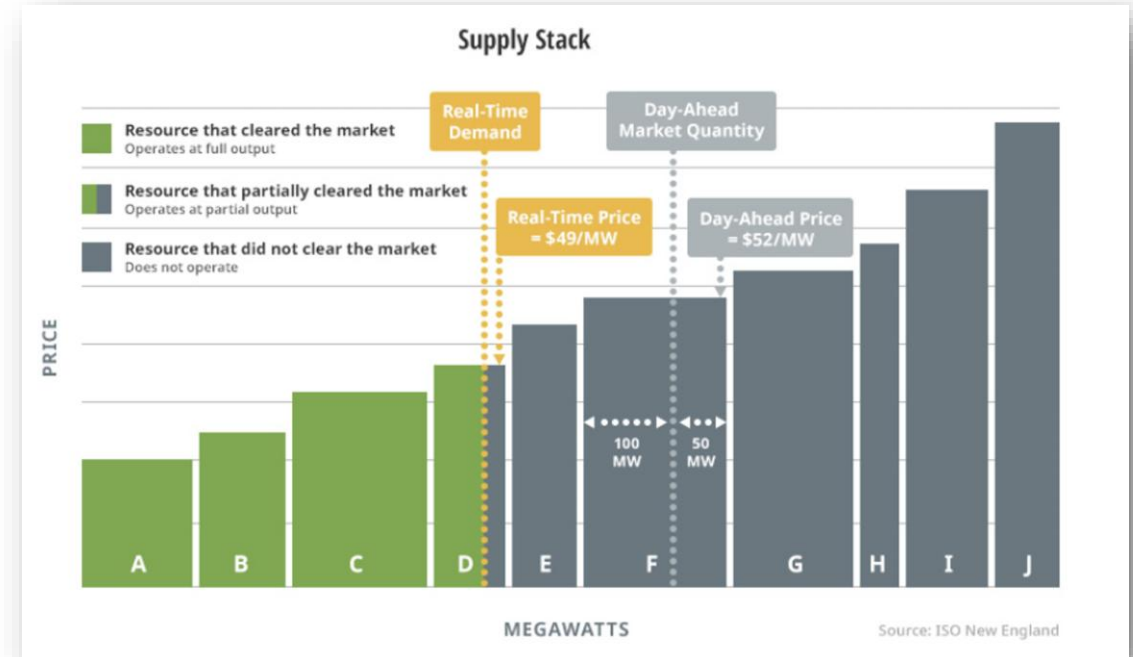
Group 13 – Power plant bidding optimization

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Introduction: Energy-Market Bidding

- Approximately $\frac{2}{3}$ of North American demand is coordinated by Independent System Operators (ISOs) and Regional Transmission Organizations (RTOs) operating competitive electricity markets
- Real-Time Market (RAM): balance supply and demand as actual conditions unfold
- **Day-Ahead Market (DAM):** participants submit for the next day with hourly resolution



<https://www.iso-ne.com/about/what-we-do/how-resources-are-selected-and-prices-are-set>

Our Partner - Enverus

Enverus is an energy data and analytics company that provides information for participants in North American electricity and commodity markets

Goal: Develop a model that uses Enverus' proprietary datasets to optimize strategies in the energy-market bidding landscape by integrating machine learning and game theory



Problem Description

Introduce Problem

Problem Description

Probabilistic Description and Uncertainty

- Current Uncertainty
 - Lack competitors' bidding strategies
 - Market Clearing Price
- Markov Property:
 - State and action space distribution

Formal Problem Description: **"Generate optimal bidding strategy with partial information."**

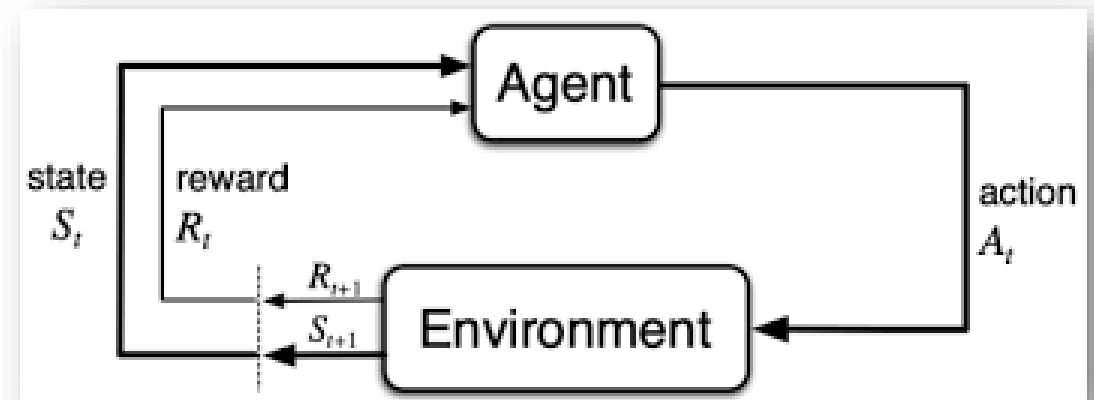
$$P(s_{t+1}, R_t | s_t, a_t) = P(s_{t+1}, R_t | s_t, a_t, s_{t-1}, a_{t-1}, \dots)$$

Reinforcement Learning Application

- Current market participants mostly rely on fixed-rule strategies and heuristic playbooks
- Fail to catch seasonal pattern changes, new renewables, increase in volatility, change in competitor strategies

Why reinforcement learning?

- Learns optimal decisions directly from historical price and load trajectories
- Can incorporate high-dimensional states
- Adapts to changing market conditions with additional data



[2] Lilian Weng. Exploration strategies in deep reinforcement learning.

TBL Analysis

People (Social)

- Supports more reliable grid operation
- Reduces sudden price spikes for users
- Helps keep electricity bills predictable

Planet (Environmental)

- Uses generation on the system efficiently
- Cuts waste from over or under-production
- Can reduce emissions with smoother dispatch

Profit (Economic)

- Improves long-run bidding performance
- Increases expected profit for generators
- Creates a more disciplined, data-driven strategy

[5] Clean Energy Wire. Setting the power price

[6] International Energy Agency. Renewables - energy system

Design Constraints

Constraints on agent and model design

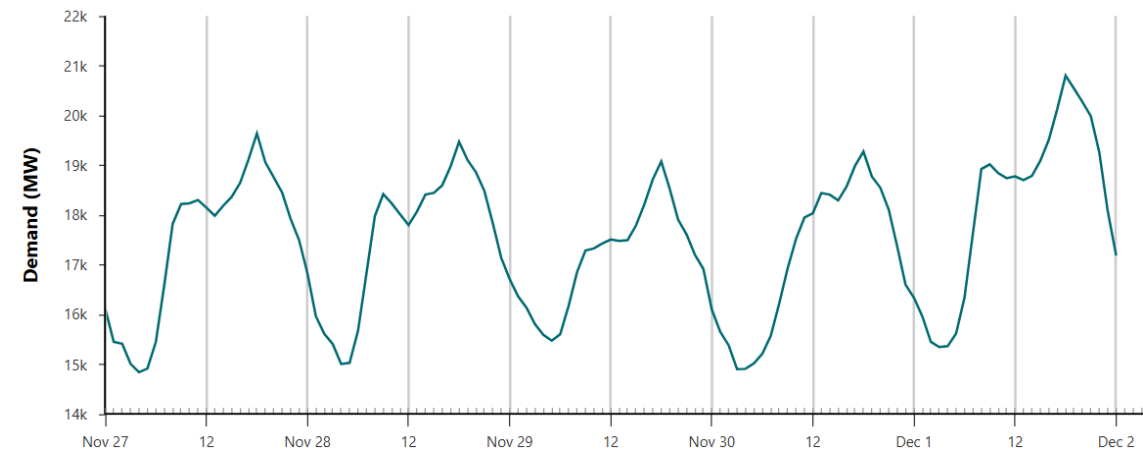
Design Constraints

Mathematical Constraints

- Operational Capacity
 - Maximum Risk
 - Financial Binding Bids
1. **Capacity constraint:** $x_{i,j} \leq X$, where $X = \text{Operational capacity}$
 2. **Risk constraint:** $\sigma_{MAX} = R_{\max} \times \sigma \geq \sigma_{ij} = P_{ij} - \mu$

Computational Constraints

- Historical and projected time series data
- Model calculation time
- Model computational requirements



Electricity demand time series representation

<https://www.ieso.ca/power-data>

Markov Decision Process (MDP) Formulation

States, Actions, Rewards

State Space $\mathcal{S}$

- L_t^{hist} : Historical market loads (MWh)
- $\hat{L}_t^{\text{DA}}$: Forecasted market loads (MWh)
- λ_t^{hist} : Historical locational marginal pricing (LMP) (\$/MWh)
- $t \bmod 24$: Hour of the day
- Discrete state space

$$\mathcal{S} = \mathcal{L}^k \times \hat{\mathcal{L}} \times \Lambda^k \times \{0, 1, \dots, 23\}$$

Action Space $\mathcal{A}$

- $\mathcal{P} = \{p_0, p_1, \dots, p_{N_P-1}\}$
- $\mathcal{Q} = \{q_0, q_1, \dots, q_{N_Q-1}\}$
- $a_t = (p_i, q_j), p_i \in \mathcal{P}, q_j \in \mathcal{Q}$

$$\mathcal{A} = \mathcal{P} \times \mathcal{Q}$$

Reward Function $\mathcal{R}$

- λ_{t+24}^{DA} : realized day-ahead market clearing price
- c^{gen} = generator marginal cost: cost of generating energy
- $Q_t^{\text{cleared}} = f_t q_j$
- $R_t \equiv R(s_t, a_t) = (\lambda_{t+24}^{DA} - c) Q_t^{\text{cleared}}$

$$f_t = \begin{cases} 1, & p_i < \lambda_{t+24}^{DA} \\ 0, & p_i > \lambda_{t+24}^{DA} \\ U[0, 1], & p_i = \lambda_{t+24}^{DA} \end{cases}$$

Environment Simulator

Hidden Environment, Market Dynamics

Environment Simulator

Hidden Environment

- Simulation of the day-ahead market built from real historical LMP and load data sourced from Enverus and EIA
- Creates a learning setting where the agent trains on historical market behavior using only information a real bidder would have

Episodes

- One episode = one day
- Simulator partitions past historical day for LMP and realized load
- Steps through hours 0 to 23
- Agent chooses action
- Applies clearing rules of environment
- Updates state for the next hour

Market Dynamics

Market Transitions

- (s_t, a_t) determines the bid's clearing outcome producing reward R_t
- The market evolves deterministically from the historical LMP and load, giving s_{t+1}
- These dynamics define a transition structure and form a controlled stochastic system
- $(s_t, a_t) \rightarrow$ (whether the bid clears, profit earned, state for the next hour)

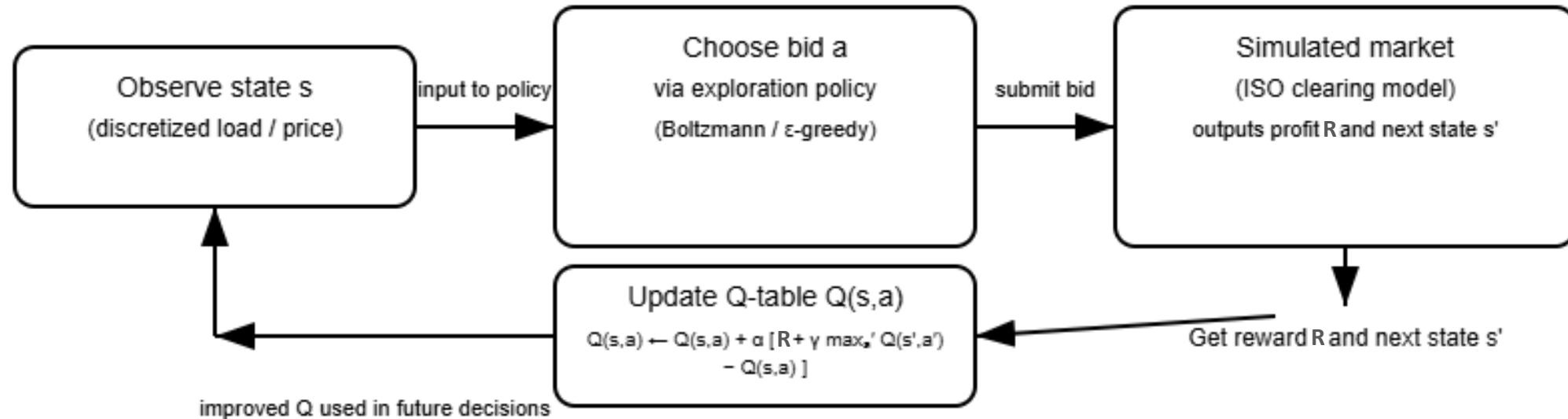
Profit/Reward Formulation

- Generation Cost (C^{gen})
:= cost of producing energy based on asset parameters
- Revenue ($r_{d,t}^{\text{rev}}$): $= Q_t^{\text{cleared}} * \lambda_{t+24}^{DA}$
- Reward ($R_{d,t}$): $= r_{d,t}^{\text{rev}} - C^{\text{gen}}$
- Daily Return (R_d): $= \sum_{t=0}^{23} R_{d,t}$

Q-Learning Approach

Tabular Q-Learning and Exploration Strategy

Q-Learning Architecture



Update Rule

- $Q(s, a)$'s start at 0
- Observe effects of many trials
- Converge toward Bellman-consistent estimates

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[R + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

How The Agent Explores Actions

- Boltzmann is primary policy
- ϵ -greedy also implemented as a baseline
- Action probabilities $\propto \exp\left(\frac{Q(s, a)}{\tau}\right)$

$$\pi(a|s) = \frac{\exp(Q(s, a)/\tau)}{\sum_{a'} \exp(Q(s, a')/\tau)}$$

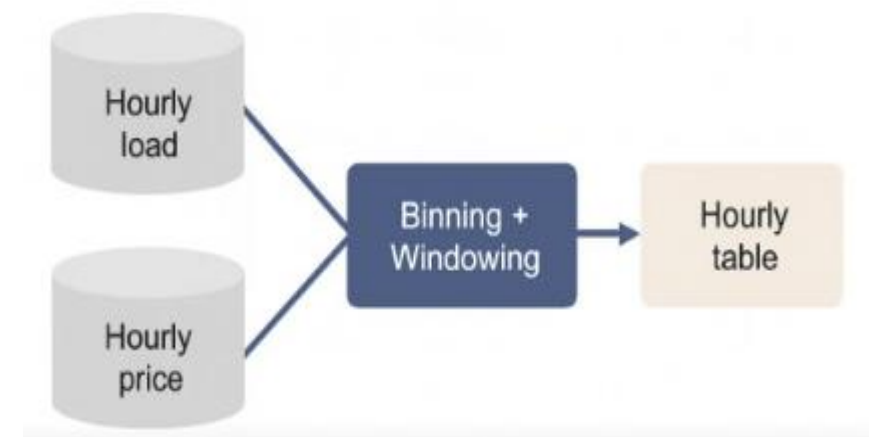
Current Progress

Model Development

Current Progress

Data ingestion

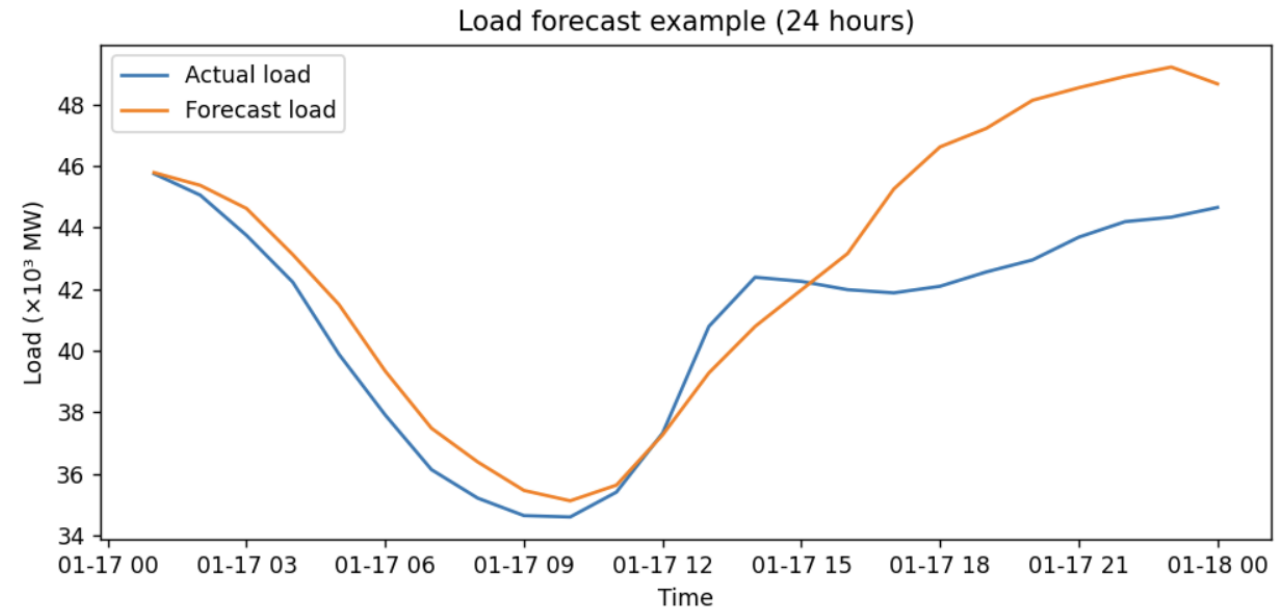
- We pull in system load, market price, and forecasted load, and line them up into one clean hourly table.
- We use a simple tool that cuts each variable into bins and picks the binning method that avoids empty or unbalanced bins.
- We build 14-day windows of this binned data that move forward one day at a time, so each run uses two weeks of history in order



Current Progress

SARIMAX module

- Fitted a SARIMAX time-series model to historical ERCOT demand over the chosen training period to generate short-term load forecasts.
- Connected these forecasts into the state so they are treated as an additional input alongside demand and price whenever they are available.

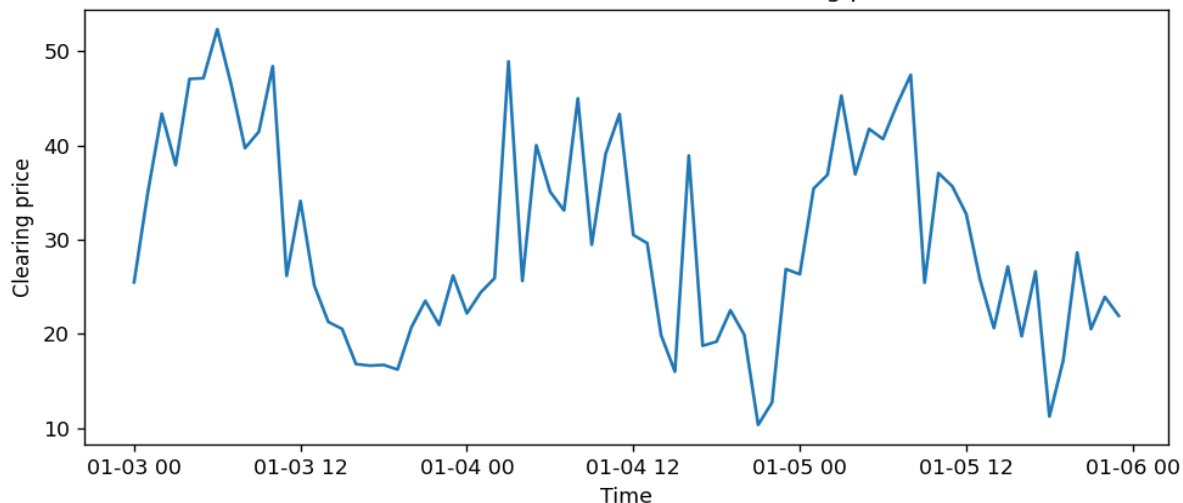


Current Progress

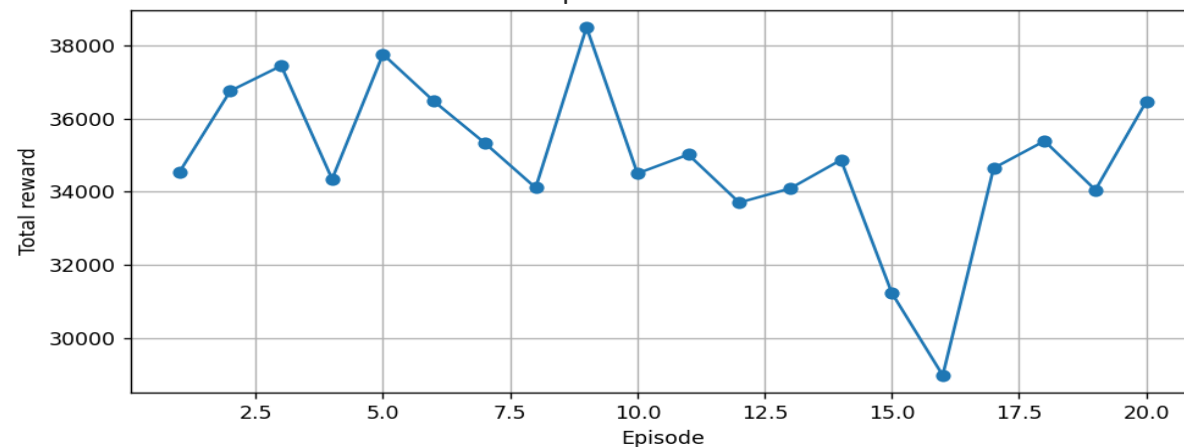
Action/State Prototypes & Q-Learning

- We represent each hour using a small set of binned features, including load, price, forecasted load, and time of day.
- We build a set of possible bids by combining different price levels with different quantity levels.
- We run a learning loop that tries actions, simulates market results, updates values over time, and saves what it learns.

Simulated environment trace - clearing price

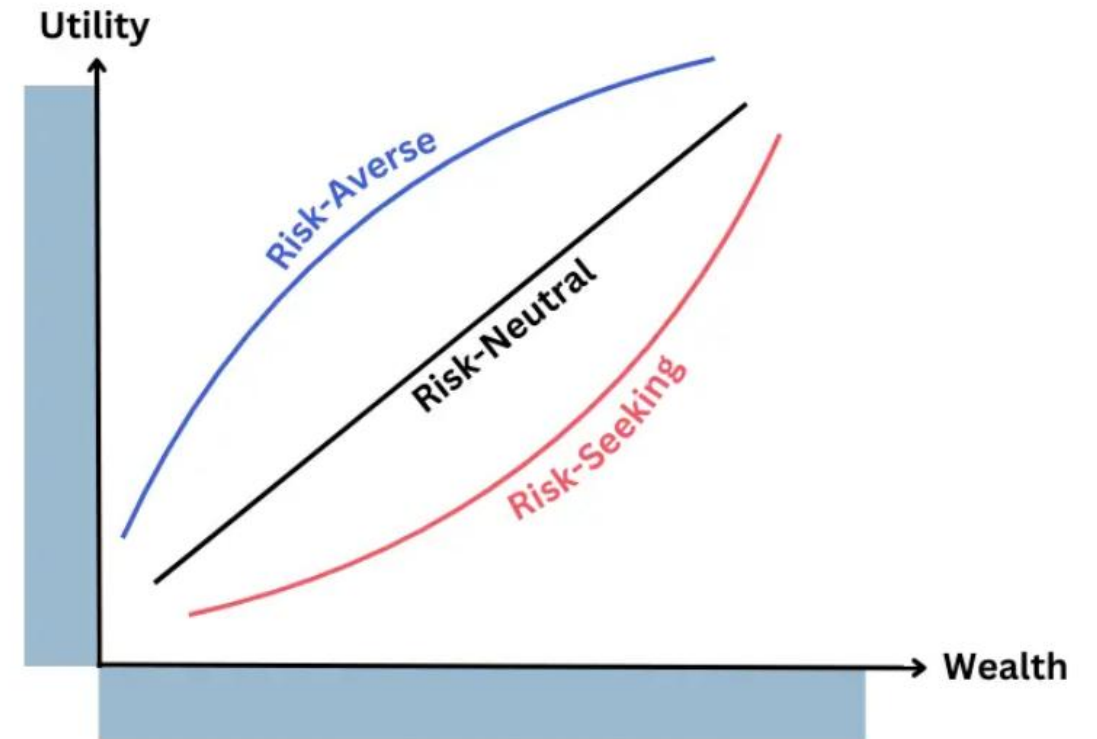


Episode Rewards



Design Challenges

- Mathematical Design: Risk Incorporation
- Computational Design: State and Action Space
- Computational Design: Data Collection



Graph of “Wealth” or Profit over “Utility, must be added within model

<https://maseconomics.com/risk-uncertainty-and-insurance-in-economics-a-comprehensive-guide/>

Next Steps

Upcoming Changes in Semester 2

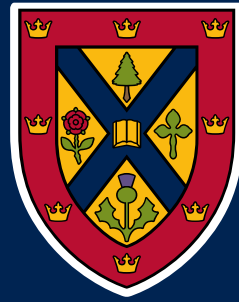
Next Steps

Model Tuning

- State and Environment Expansion
- Hyperparameter Tuning
- Maximum Risk Constraint

Game Theoretic

- Capturing competitor behavior with game theory
- Conversion to Q-Learning with Equilibrium



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